



Solar Box Dryer

Zenitek
Towards Sustainable Future

For Entrepreneur's



YOUTUBE: zeniteksolar

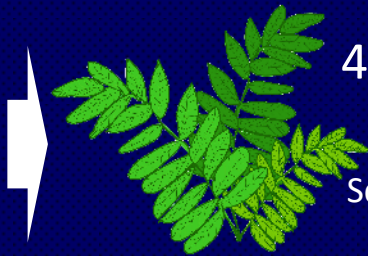
Call us +91- 8903852623



Structural and Material Specifications

- **Structure:** GI steel frame
- **Box Size:** 4 ft (L) × 4 ft (W) × 4 ft (H)
- **Leg Height:** 8 inches
- **Solar Collector:** 6 mm double-walled polycarbonate sheet with UV protection coating on both sides
- **Max Temperature:** 55 to 60°C (sunny day)
- **Tray Material:** Stainless steel 304, food-grade perforated sheet
- **Tray Area (per unit):** 6.25 Sq.Ft
- **Number of Trays:** 4 (default) + 4 (optional)
- **Door:** Concealed hinges with magnetic latches

Holding capacity
(Wet weight)



400 grams
Per
Square foot

Lighter

2.5 Kg
Per
Square foot



(Example)

Denser



Electrical Heater Unit

- A 500 W, 220 V AC strip-type electrical heater is provided as the default heater for night-time operation to keep the drying chamber warm. Higher wattage heaters can be provided **optionally (upto 3 kW)**, if increased heating capacity is required.
- A 4-inch AC fan is used as a blower for air circulation. Temperature is controlled by a **thermostat**, and power supply is managed through an **MCB** with **indicator lamps** for power and operation status.



Air Circulation Arrangement

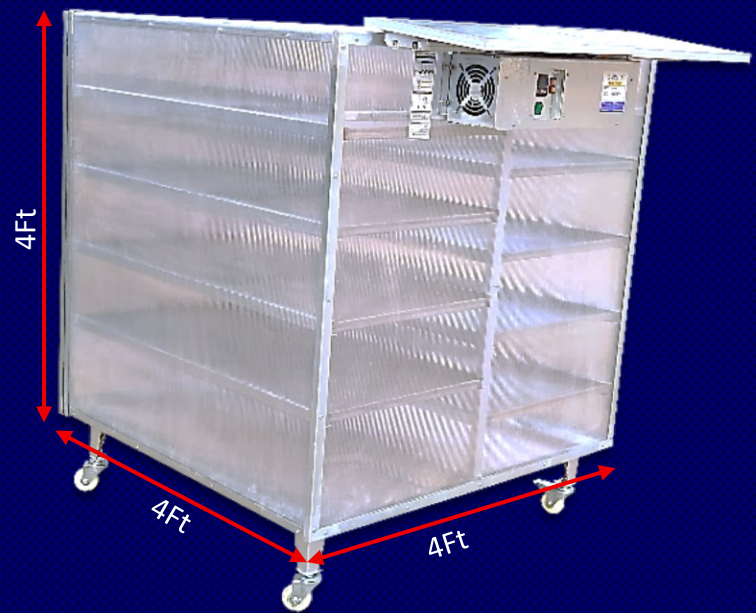
A 4-inch AC fan is mounted at the top to push hot air downwards through the drying chamber. The trays are arranged in a stair-step (stepped) pattern, allowing better air distribution and improved airflow across all trays

Exhaust Unit

Two 4-inch, 24 V DC fans are used to exhaust hot and moist air from the system, helping to maintain optimal drying conditions

Mobility Unit

Four 3-inch PU caster wheels are provided for easy movement of the system. Two wheels are fitted with locking mechanisms to keep the unit stable during operation.



Control Unit

- **Temperature Controller**
Shows the inside temperature and turns the exhaust fan ON when the temperature goes above the set value.
- **Timer Controller**
Turns the exhaust fan ON and OFF at set time intervals. The ON and OFF time can be adjusted up to 10 minutes, which is useful on cloudy days to remove moist air.



Power Supply Unit

- The control unit is powered by a **24 V, 20 W solar PV panel** through a Victron MPPT charge controller. A **24 V, 7 Ah battery** is provided only to handle short-term solar power fluctuations and ensure stable operation during varying sunlight conditions. The battery is not intended for night-time operation.
- The Victron MPPT supports mobile interaction via the Victron Connect app, allowing real-time monitoring of solar input, battery status, and system performance





Price (including GST)

4 tray ₹55,000/-

8 tray ₹65,000/-

Packing and Transportation

Local Area:

The dryer will be transported directly in a vehicle, and no packing is required.

Outstation:

The dryer will be packed in a welded cubical steel frame covered with plywood for safe transportation.

Packing charges: ₹8,000/-

For both local and outstation deliveries, the transportation cost is on a **To-pay basis.**



Installation

Installation is **DIY (Do It Yourself)**. The legs, solar PV panel, and control panel are provided separately and can be easily assembled at the site.